

FaciliTrak

AI-Powered Classroom & Facility
Condition Reporting System

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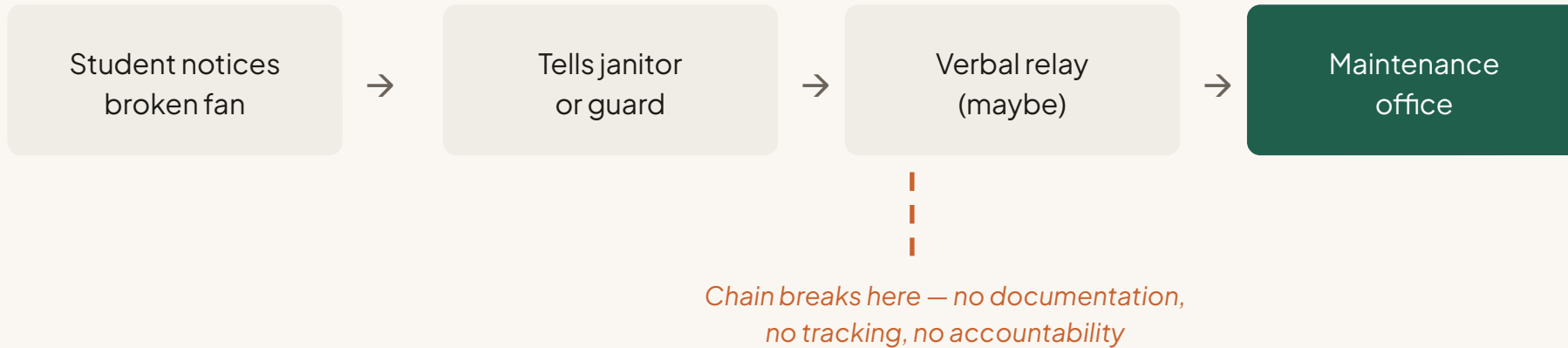
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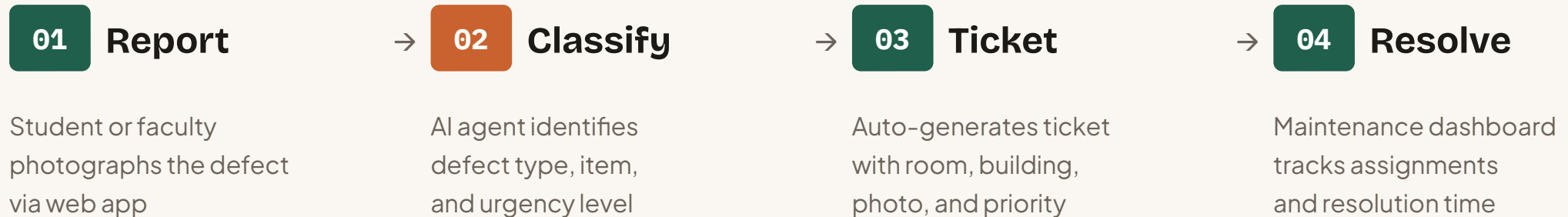
THE PROBLEM

Broken facilities persist for weeks because reports never reach the right people.



THE SOLUTION

FaciliTrak replaces the verbal chain with a structured, AI-assisted reporting pipeline.



What makes FaciliTrak different

AI-powered classification

No manual tagging — the agent classifies defect type and urgency from the photo

Structured accountability

Every report is a trackable ticket with timestamps, assignments, and resolution verification

Recurrence detection

Flags rooms with repeated issues of the same type, surfacing systemic maintenance failures

OBJECTIVES

General Objective

To design and develop an AI-powered web-based system that streamlines the reporting, classification, tracking, and resolution of classroom and facility maintenance issues in Philippine educational institutions.

Specific Objectives

- SO₁** To develop an AI-based defect classification module that categorizes reported issues by type (electrical, plumbing, structural, furniture, HVAC, sanitation) and estimates urgency level.
- SO₂** To design and implement a structured maintenance ticketing workflow with automated ticket generation, assignment routing, and resolution tracking.
- SO₃** To build a real-time analytics dashboard with building-level heatmaps for identifying recurring problem areas and monitoring mean time-to-resolution.
- SO₄** To implement role-based access control via Supabase RLS enforcing reporter, maintenance staff, and administrator permission boundaries.
- SO₅** To evaluate the system's usability, defect classification accuracy, and impact on maintenance response time using ISO/IEC 25010 and comparative pre/post-deployment analysis.

SCOPE OF THE PROJECT

Core Features

- Photo-based defect reporting with geolocation tagging
- AI agent classification (6 defect types + urgency scoring)
- Automated ticket generation with room/building metadata
- Resolution workflow: Open → Assigned → In-Progress → Resolved → Verified
- Recurrence detection for same-room, same-defect patterns
- Analytics dashboard with building heatmaps and KPI tracking
- Real-time notifications for status changes

User Access Levels

Reporter

Students & faculty — submit reports, track own tickets, verify resolutions

Maintenance Staff

Receive assignments, update status, log work completed, attach evidence

Administrator

Full dashboard access, manage staff, view analytics, configure buildings/rooms

LIMITATIONS

Explicitly out of scope — Phase 2 candidates

IN SCOPE

- ✓ Web-based responsive application (mobile browser access)
- ✓ AI defect classification for 6 predefined categories
- ✓ Single-institution deployment and testing
- ✓ Ticketing and resolution workflow (5 stages)
- ✓ Role-based access for 3 user types
- ✓ Requires active internet connection

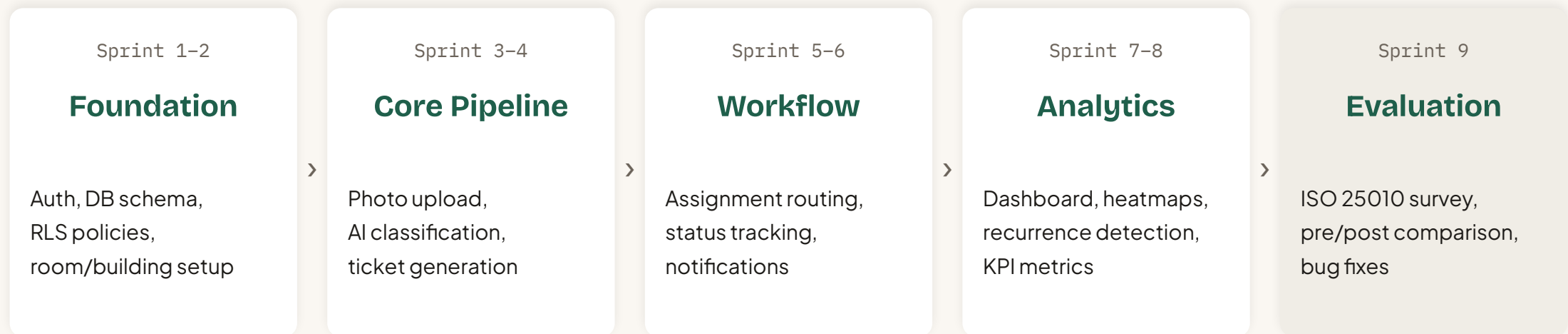
OUT OF SCOPE (Phase 2)

- ✗ Native mobile application (iOS / Android)
- ✗ Offline reporting and sync
- ✗ Multi-institution / district-wide deployment
- ✗ Automated work order dispatch to external contractors
- ✗ Integration with existing school MIS or DepEd systems
- ✗ Predictive maintenance analytics (ML-based forecasting)

Setting clear boundaries protects the project from scope creep and demonstrates feasibility awareness. Each out-of-scope item is a documented Phase 2 candidate.

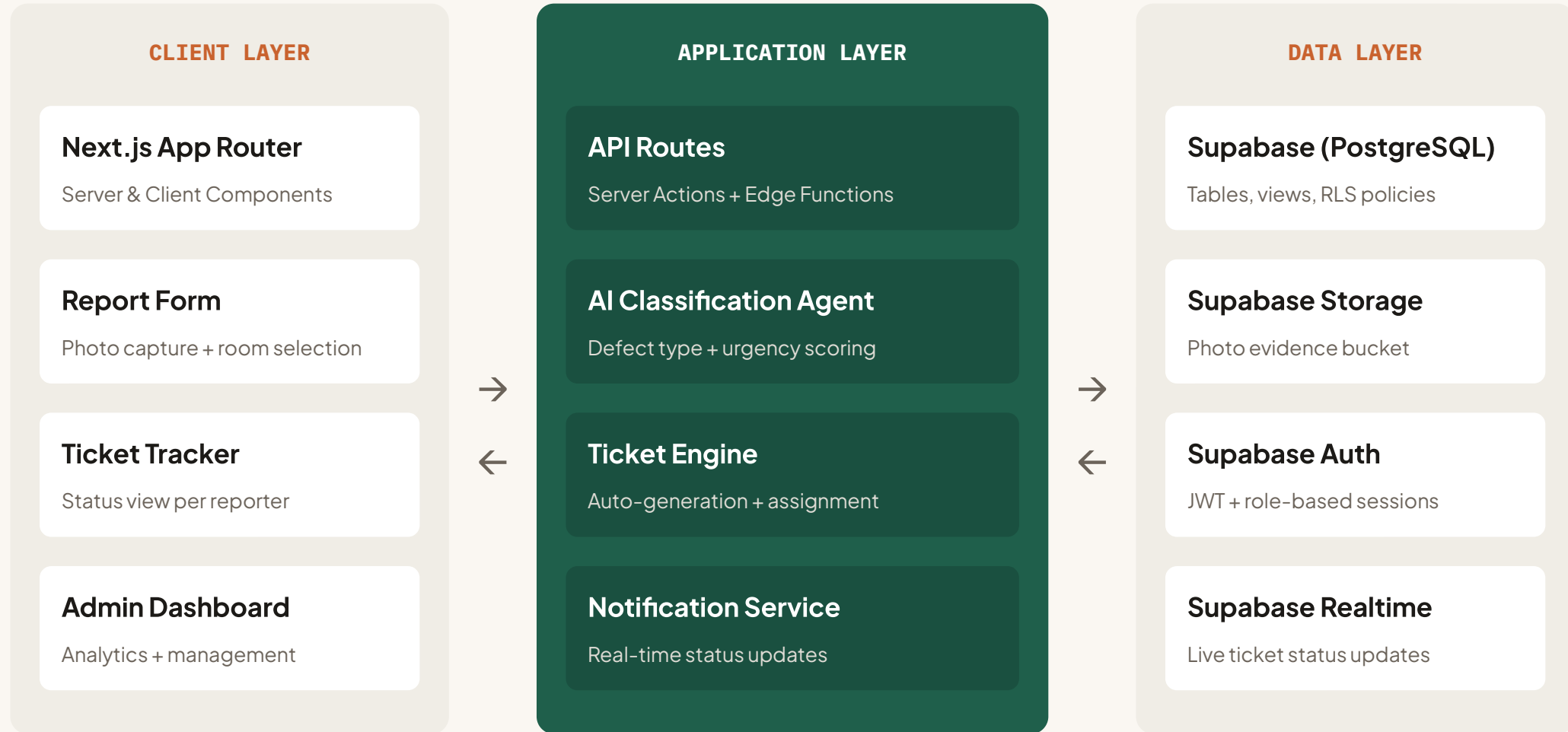
Agile-Scrum with 2-week sprints

Requirements will evolve after initial user feedback from reporters and maintenance staff. Agile accommodates this with iterative delivery — each sprint produces a testable increment.



[[TBD: exact start/end dates]] · ~18 weeks total · 9 sprints × 2 weeks

SYSTEM ARCHITECTURE



TECHNOLOGY STACK

Frontend

Next.js 15	App Router, Server Components
TypeScript	Strict mode, Zod validation
Tailwind CSS	Utility-first styling
shadcn/ui	Accessible component primitives

Backend & Database

Supabase	PostgreSQL + Auth + Storage + Realtime
Edge Functions	Serverless compute for AI pipeline
Row Level Security	DB-enforced access control
Zod	Runtime schema validation

AI / Classification

[[TBD: Model]]	Defect image classification
Structured Output	JSON schema for defect taxonomy
Urgency Algorithm	Safety hazard vs. inconvenience

Infrastructure & Tools

Vercel	Deployment + edge network
Git / GitHub	Version control + CI/CD
ESLint + Prettier	Code quality enforcement

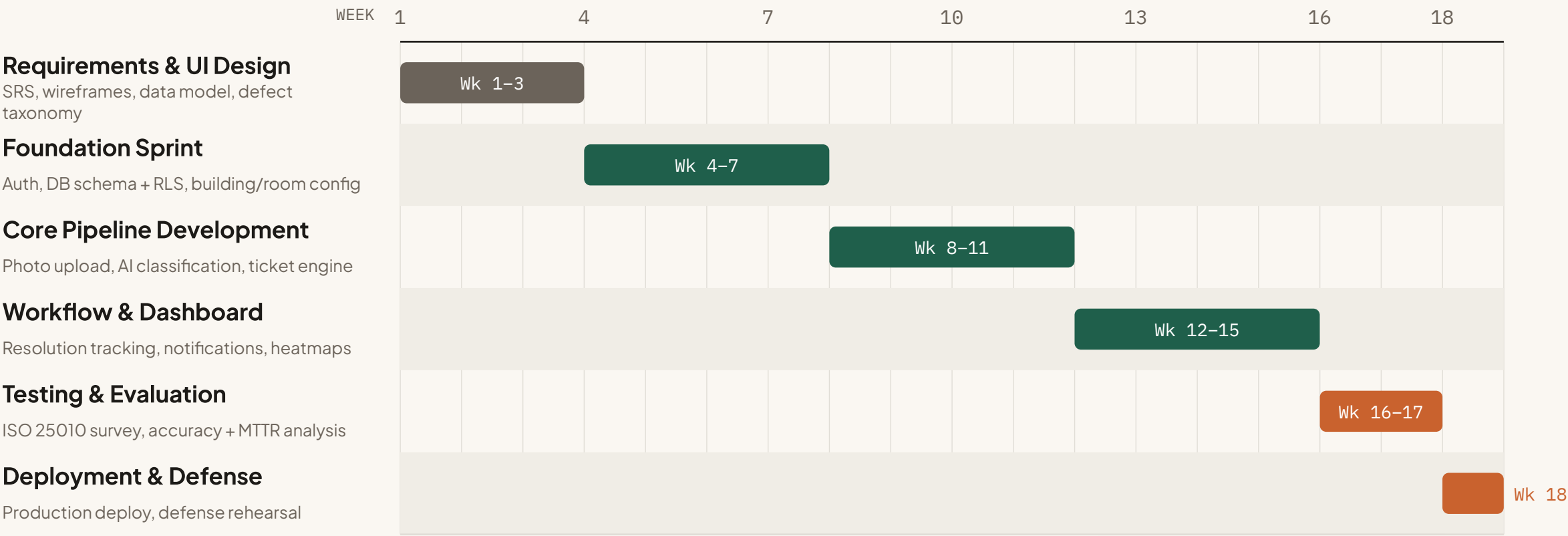
Alignment Matrix — how we prove it works

Objective	Feature	How Evaluated	Metric
SO ₁ — AI classification accuracy	Classification module	Compare AI output vs. maintenance supervisor assessment	Classification accuracy rate (%)
SO ₂ — Ticketing workflow	Ticket engine + resolution pipeline	Mean time-to-resolution pre/post deployment	MTTR reduction (days)
SO ₃ — Analytics dashboard	Heatmaps + KPI tracking	Reporting volume increase measurement	# of reports pre vs. post deployment
SO ₄ — Role-based access control	Supabase RLS policies	Penetration testing + RLS audit	Zero unauthorized access violations
SO ₅ — Usability evaluation	Entire system	ISO/IEC 25010 survey (n = [[TBD]])	Weighted mean ≥ 4.21 (Highly Acceptable)

Instrument: ISO/IEC 25010 adapted questionnaire · Respondents: [[TBD: students, faculty, maintenance staff]] · Sample size: [[TBD: justified via Slovin's formula]] · Analysis: Weighted mean with 5–point Likert scale

IMPLEMENTATION ROADMAP

18-week development timeline across six phases



[[TBD: exact start/end dates]] · Total: 18 weeks · Phase boundaries align with sprint checkpoints

FaciliTrak

AI-Powered Classroom & Facility Condition Reporting System

OBJECTIVES

SO1 Develop AI defect classification (6 types + urgency)

SO2 Implement structured ticketing workflow

SO3 Build analytics dashboard with heatmaps

SO4 Enforce role-based access via Supabase RLS

SO5 Evaluate with ISO 25010 + pre/post analysis

EXPECTED OUTPUT

Working web application deployed on Vercel

AI classification module with measured accuracy

Maintenance dashboard with live ticket tracking

Pre/post deployment MTTR comparison report

ISO 25010 usability evaluation results

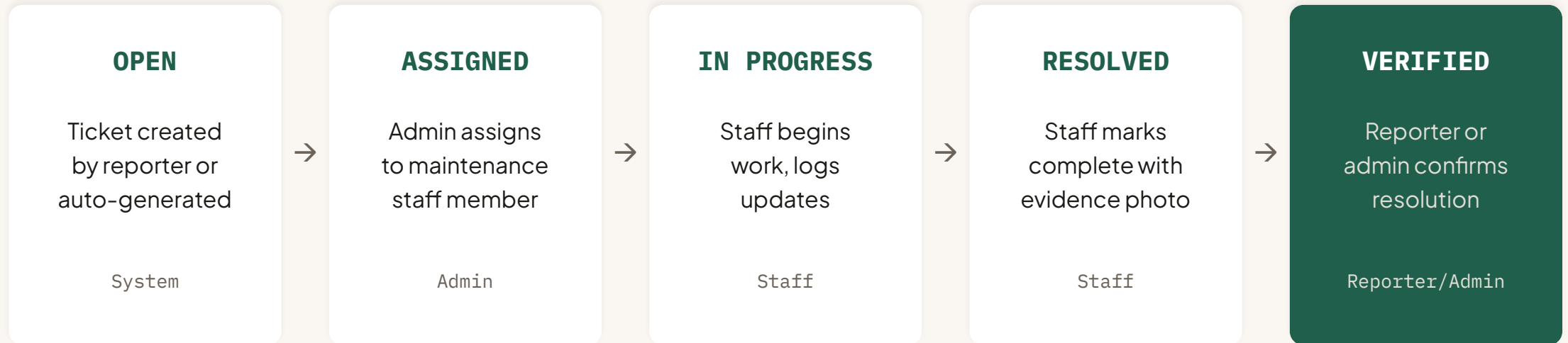
We welcome your questions and feedback.

APPENDIX A1 – DEFECT TAXONOMY

Defect Type	Example Items	Default Urgency	Code
Electrical	Non-functional outlets, exposed wiring, broken switches, flickering lights	High (safety hazard)	ELEC
Plumbing	Leaking faucets, clogged drains, broken toilets, water damage	Medium-High	PLMB
Structural	Leaking roofs, cracked walls, broken windows, damaged floors	High (safety hazard)	STRC
Furniture	Broken chairs, damaged desks, unstable tables, broken cabinet locks	Medium (inconvenience)	FURN
HVAC	Busted ceiling fans, broken AC units, poor ventilation	Low-Medium	HVAC
Sanitation	Overflowing trash, damaged waste bins, pest issues, mold	Medium (health)	SNTN

Urgency scoring: Safety hazards (exposed wiring, structural cracks) are auto-escalated to HIGH. The AI agent can override default urgency based on visual severity assessment from the photo.

APPENDIX A2 – RESOLUTION WORKFLOW



Rejection path: If verification fails, ticket returns to ASSIGNED status with a rejection note.

Escalation: Tickets unresolved after [[TBD]] days auto-escalate to admin notification.

RA 10173 (Data Privacy Act of 2012) Compliance

What data is collected?

User name, email, institutional role, defect photos, room/building location. No biometric or sensitive personal data.

Where is data stored?

Supabase cloud infrastructure (AWS region). Photos in Supabase Storage with access-controlled URLs.

Who can access what?

Enforced via Supabase RLS: reporters see only their own tickets; staff see assigned tickets; admins see all. No client-side access checks.

Retention policy

Active tickets retained indefinitely for analytics. Resolved tickets archived after [[TBD]] months. Photos retained with tickets.

Consent mechanism

Explicit consent during registration. Users can request data deletion per RA 10173 rights.

APPENDIX A4 – EXISTING SOLUTIONS

Criteria	Manual Process (Current)	SchoolDude / FMX	Google Forms + Sheets	FaciliTrak (Proposed)
Structured reporting	✗	✓	Partial	✓
AI classification	✗	✗	✗	✓
Photo evidence	✗	✓	✓	✓
Urgency scoring	✗	Manual	✗	AI-automated
Recurrence detection	✗	✗	✗	✓
Filipino school context	—	✗ (US-centric)	Neutral	✓ (designed for)
Cost	Free	\$\$\$ / license	Free	Free (Supabase free tier)
Resolution tracking	✗	✓	✗	✓ (5-stage)

Honest comparison: SchoolDude/FMX offers superior resolution tracking out of the box. FaciliTrak's advantages are AI classification, Filipino school context, and zero licensing cost.